KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Year 10 Yearly Test Help Sheet

Year 10 Transition · 2026 · Key ideas, formulas and worked examples for every topic

Use this help sheet while you revise. Each box (A1, A2, …) matches the questions in the revision booklet.

A · Financial mathematics

A1 Percentage of an amount

Change the % into a decimal (divide by 100), then multiply.

> **Example:** 28% of \$50 → 0.28 × 50 = **\$14**

A2 GST (10%)

- Add GST: price × 1.1
- GST in a price that already has GST (GST inc.): price ÷ 11
- Price before GST: price ÷ 1.1

> **Example:** \$55 (GST inc.): GST = 55 ÷ 11 = **\$5**

A3 Discounts

Discount = price × rate. New price = price − discount.

> **Example:** \$60 with 20% off: 0.2 × 60 = \$12, 60 − 12 = **\$48**

A4 Simple interest

Formula
$I = P \times R \times N$

P = amount invested or borrowed, R = rate as a decimal (5% → 0.05), N = number of years. Total value = $P + I$.

> **Example:** \$400 at 5% for 2 years: 400 × 0.05 × 2 = **\$40**

A5 Increase or decrease with a multiplier

- Increase by 8% → × 1.08
- Decrease by 8% → × 0.92 (100% − 8% = 92%)

> **Example:** Increase \$50 by 10%: 50 × 1.10 = **\$55**

A6 Mark-up and profit

- Cost price = what the shop pays. Selling price = what the customer pays.
- Mark-up of 30% → cost × 1.3
- Profit = selling price − cost price

> **Example:** Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = 12 − 10 = **\$2**

A7 Wages and overtime

- Pay = hours × hourly rate
- Time and a half = 1.5 × hourly rate
- Double time = 2 × hourly rate

> **Example:** \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = **\$90**

A8 Commission

Commission = rate (as a decimal) × sales.

> **Example:** 4% of \$900 sales: 0.04 × 900 = **\$36**

A9 Gross pay and net pay

- **Gross pay** = total pay before anything is taken out (hours × rate)
- **Net pay** = gross pay − tax − super (what you take home)

> **Example:** Gross \$800, tax \$120, super \$40: 800 − 120 − 40 = **\$640**

A10 Loans and payment plans

- Change years into months: 1 year = 12 months
- Total repaid = repayment × number of months
- Interest = total repaid − amount borrowed

> **Example:** Borrow \$500, repay \$50 a month for 1 year: 50 × 12 = \$600, interest = 600 − 500 = **\$100**

A11 Compound interest

Formula
$A = P(1 + R)^n$ A = final value, P = amount invested, R = rate as a decimal, n = number of years

> **Example:** \$2000 at 5% for 2 years: $A = 2000 \times 1.05^2$ = **\$2205**. Interest = 2205 − 2000 = **\$205**

> **Watch out:** 5% compound interest means × 1.05 each year. Calculator: type 2000 × 1.05 ^ 2 =. Interest = final value − amount invested.

B · Measurement

B1 Converting units

Length, capacity and mass
1 cm = 10 mm 1 m = 100 cm 1 km = 1000 m
1 L = 1000 mL 1 kL = 1000 L 1 kg = 1000 g

Big unit → small unit: **multiply**. Small unit → big unit: **divide**.

> **Example:** 5.3 km → m: 5.3 × 1000 = **5300 m**.
> 2500 mL → L: 2500 ÷ 1000 = **2.5 L**

B2 Scientific notation

A number between 1 and 10, times a power of 10. Count how many places the decimal point moves.

> **Example:** 2 350 000 = **2.35×10^6** (the point moves 6 places)

> **Example:** $4.1 \times 10^3 = 4.1 \times 1000 =$ **4100**

B3 Perimeter

Perimeter = add up the lengths of all the sides (the distance around the outside). Units: m, cm.

> **Example:** Rectangle 12 m by 7 m: 12 + 7 + 12 + 7 = **38 m**

B4 Circles

Formulas
Circumference $C = 2 \times \pi \times r$ Area $A = \pi \times r \times r$

r = radius (centre to edge). If you are given the diameter, halve it first.

> **Example:** Radius 5 cm: C = 2 × π × 5 = **31.4 cm**, A = π × 5 × 5 = **78.5 cm²**

B5 Area

Shape	Area
Rectangle	length × width
Triangle	½ × base × height
Trapezium	½ × (top + bottom) × height

Top and bottom are the parallel sides. The height makes a right angle with the base. Area units are squared: cm², m².

> **Example:** Triangle, base 8 cm, height 5 cm: ½ × 8 × 5 = **20 cm²**

B6 Volume of cubes and boxes

Formulas
Cube: V = side × side × side
Box (rectangular prism): V = length × width × height

Volume units are cubed: cm³, m³.

> **Example:** Box 4 cm × 3 cm × 2 cm: 4 × 3 × 2 = **24 cm³**

B7 Surface area of a box

A box has 6 faces in 3 matching pairs: top and bottom, front and back, the two ends. Find the area of one face of each pair, double it, then add.

> **Example:** Box 10 × 4 × 2: 2 × 40 + 2 × 20 + 2 × 8 = 80 + 40 + 16 = **136 cm²**

B8 Volume of a cylinder

Formula
$V = \pi \times r \times r \times h$

1 cm³ holds 1 mL, so the volume in cm³ is the capacity in mL.

> **Example:** r = 3 cm, h = 10 cm: π × 3 × 3 × 10 = **282.7 cm³** = 282.7 mL

Watch out: Round to 1 decimal place only at the end. Always write the units: cm for length, cm² for area, cm³ for volume.

C · Statistics and probability

C1 Types of data

Categorical (words or groups)	
Nominal	no order: eye colour, type of pet
Ordinal	has an order: small, medium, large
Numerical (numbers)	
Discrete	counted: number of goals
Continuous	measured: height, time, mass

C2 Mean, median, mode and range

- **Mean** = add up the numbers, then divide by how many numbers there are
- **Median** = the middle number when they are in order
- **Mode** = the most common number
- **Range** = biggest − smallest

> **Example:** 3, 8, 1, 8, 5 → in order 1, 3, 5, 8, 8. Mean = 25 ÷ 5 = **5**, median = **5**, mode = **8**, range = 8 − 1 = **7**

C2 Reading a stem-and-leaf plot

The stem is the tens digit. The leaf is the units digit. The key tells you how to read it.

This plot shows 24, 29, 31, 31, 37. Lowest = 24, highest = 37, mode = 31.

Stem	Leaf	3 | 1 means 31
2	4 9	
3	1 1 7	

C3 Probability

Formula

Probability = $\frac{\text{number of ways it can happen}}{\text{total number of ways}}$

Write the answer as a fraction and simplify it if you can.

> **Example:** A letter from BOOK: P(O) = $\frac{2}{4} = \frac{1}{2}$.
> Dice, an even number (2, 4, 6): $\frac{3}{6} = \frac{1}{2}$

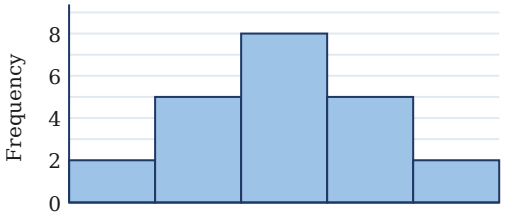
C4 Time-series graphs

- **Upward trend**: the line goes up over time
- **Downward trend**: the line goes down over time
- **No trend**: the line goes up and down with no pattern

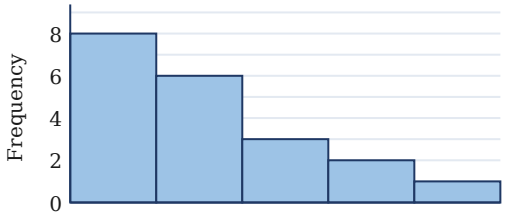
C5 Frequency tables, graphs and shape

- Total = add up the frequency column
- Percentage = frequency ÷ total × 100
- Column graph: frequency up the side, groups along the bottom
- **Symmetrical**: the same shape on both sides of the middle. **Skewed**: bunched at one end with a long tail

> **Example:** 9 out of 20 students: 9 ÷ 20 × 100 = **45%**



Symmetrical



Skewed

D · Other Year 10 topics

D1 Index laws

- When you **multiply**, add the powers: $m^2 \times m^6 = m^8$
- When you **divide**, subtract the powers: $m^7 \div m^2 = m^5$
- x on its own means x^1

D2 Substitution

Swap the letter for its number. $5a$ means $5 \times a$. Multiply before you add or subtract.

> **Example:** If $a = 3$: $2a + 1 = 2 \times 3 + 1 = \mathbf{7}$

D3 Solving equations

Do the opposite to both sides, one step at a time.

> **Example:** $3x + 1 = 13$ → take away 1: $3x = 12$ → divide by 3: $\boldsymbol{x = 4}$

> **Example:** $\frac{x + 1}{2} = 4$ → times by 2: $x + 1 = 8$ → take away 1: $\boldsymbol{x = 7}$

Check: put your answer back into the equation.

D4 Straight-line graphs

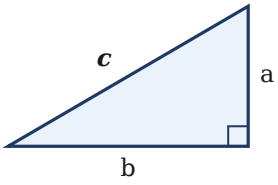
y* = *mx* + *c
m is the gradient (how steep the line is). *c* is the *y*-intercept (where the line crosses the *y*-axis).

Example: For $y = 3x + 2$, when $x = 4$: $y = 3 \times 4 + 2 = \mathbf{14}$. Gradient = 3, *y*-intercept = 2

To plot (4, 9): go across to 4, then up to 9. Put a dot. Join the dots with a ruler.

D5 Pythagoras' theorem

Formula
$c^2 = a^2 + b^2$ (*c* is the hypotenuse, the longest side, opposite the right angle)



Example: $a = 3$, $b = 4$: $c^2 = 9 + 16 = 25$, $c = \sqrt{25} = \mathbf{5}$

D6 Trigonometry: tan

Formula
tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

Example: Angle 20°, adjacent 5 cm: $x = 5 \times \tan 20° = \mathbf{1.8\ cm}$

Watch out: The opposite side is across from the angle. The adjacent side is next to the angle (not the hypotenuse). Check your calculator shows D or DEG.